SEMESTER II

23-200-0201B LINEAR ALGEBRA & TRANSFORM TECHNIQUES

Course Outcomes:

On completion of this course the student will be able to:

1. Solve linear systems of equations and to determine Eigen values and vectors of a matrix.
2. Exemplify the concept of vector space and subspace.
3. Determine Fourier series expansion of functions and transform.
4. Solve linear differential equation and integral equation using Laplace transform.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1										
CO3	3	3	3									
CO4	3	3	2									

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Linear Algebra 1: Rank of a matrix, solution of linear system of equations- existence, uniqueness, general form-Eigen values and Eigen vectors- properties of Eigen values - Diagonalization of a matrix - Cayley Hamilton theorem (without proof) Verification-Finding inverse and power of a matrix using it-Quadratic form-orthogonal reduction of quadratic form to Canonical form.

Module II

Linear Algebra 2: Vector space-subspace-Linear dependence and independence-Spanning of a subspace- Basis and Dimension. Inner product- Inner product spaces - Orthogonal and Orthonormal basis
–Gram- Schmidt Orthogonalization process. Linear Transformation.

Module III

Fourier Analysis: Periodic function, Fourier series, Functions of arbitrary period, Even and odd functions, Half Range Expansion, Harmonic analysis, Complex Fourier Series, Fourier Integrals, Fourier Cosine and Sine Transform, Fourier Transform.

Module IV

Laplace Transforms: Gamma functions and Beta function-Definition and properties, Laplace transforms. Inverse Laplace Transform, Shifting theorem, Transform of Derivative and Integrals, Solution of differential equation and integral equation using Laplace transform, Convolution, Unit step function, Second Shifting theorem, Laplace transform of periodic function.

References:

1. Erwin Kreyzig. (2010). Advanced engineering mathematics. (tenth edition). John Wiley & Sons, Hoboken, N.J
2. Grewal, B.S. (2013). Higher engineering mathematics. (forty third edition). Khanna Publishers, New Delhi.
3. Hsiung,C.Y and Mao, G. Y. (1999). Linear algebra. World Scientific, New Jersey.
4. Hoffman, K. and Kunze, R. (1971). Linear algebra. Prentice Hall of India, New Delhi.
5. Venkataraman, M. K. (1999). Linear algebra. The National Publishing Co, Chennai.

23-200-0202B ENGINEERING CHEMISTRY

Course Outcomes:

On completion of this course the student will be able to:

1. Explain the basic concepts of chemical thermodynamics, and quantum chemistry.
2. Illustrate the spectroscopic methods in characterizing materials.
3. Develop electrochemical methods to protect different metals from corrosion.
4. Interpret the chemistry of a few important engineering materials and their industrial applications.
5. Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	2									
CO2	3	2	3									
CO3	1	1	1									
CO4	1	1	1									
CO5	2	2	3									

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Chemical Thermodynamics: Fundamentals. First law of thermodynamics, Molecular interpretation of internal energy, enthalpy and entropy. Heat of reaction. Kirchoff's equations. Dependence on pressure and temperature. Gibbs-Helmholtz equation. Free energy changes and equilibrium constant. Chemical potential and fugacity. Thermodynamics of biochemical reactions.

Phase Rule: Terms involved in phase rule and examples, Application of phase rule to one component water system, Application of phase rule to two-component systems. (Simple eutectic systems).

Module II

Quantum Chemistry: Schrodinger wave equation – significance of Ψ, well behaved functions, Postulates of quantum mechanics, Application of quantum mechanics to simple systems - particle in 1 D box, normalization of wave function, Forms of hydrogen atom wave functions and the plots of these functions to explore their spatial variations, Quantum numbers.

Module III

Spectroscopy: Principles of spectroscopy and selection rules. Electronic spectroscopy. Vibrational and rotational spectroscopy of diatomic molecules. Applications.

1H NMR spectroscopy – Principle - Relation between field strength and frequency - chemical shift - spin-spin splitting - coupling constant - applications of NMR- MRI.

Module IV

Electrochemistry: Cell EMF- its measurement and applications. Nernst Equation and application, relation of e.m.f. with thermodynamic functions (ΔH, ΔF and Δ S). Lead storage battery. Corrosion; causes, effects and its prevention.

Polymers- Classifications- Thermoplastics and thermosetting plastics- A brief account of conducting polymers (polypyrrole and polythiophene) and their applications.

Lubricants- Introduction solid and liquid lubricants- Properties of lubricants-Viscosity index- flash and fire point- cloud and pour point- aniline value.

Refractories: Classifications – Properties of refractories.

Laboratory Experiments to be conducted in the virtual lab mode.

List of Experiments (Minimum six experiments shall be conducted)

1. Determination of the partition coefficient of a solute in two immiscible liquids.
2. Phase diagram of two component System (Naphthalene-diphenylamine)
3. Conductometric titration of Strong acids with Strong base.
4. Potentiometric titration: Fe2+ vs KMNO4

5. Heat of neutralization
6. Verification of Beer-Lambert's law
7. Determination of rate constant of a reaction.
8. Determination of total hardness of water by EDTA method.
9. Determination of COD of water sample.
10. Determination of alkalinity of water.
11. Determination of chloride content of water by Mohr's method.
12. Determination of dissolved oxygen in a given water sample.
13. Determination of acidity of water sample.
14. Determination of adsorption of acetic acid by charcoal.
15. Determination of acidity of water sample

References:

1. B. H. Mahan and R. J. Meyers. University Chemistry, 4th Edition, Pearson publishers. (2009).
2. Peter W. Atkins, Julio de Paula, and James Keele. Physical Chemistry, 11th Edition, Oxford publishers. (2018).
3. M. J. Sienko and R. A. Plane. Chemistry: Principles and Applications, 3rd Edition, McGraw-Hill Publishers. (1980).
4. C. N. Banwell. Fundamentals of Molecular Spectroscopy, 5th Edition, McGraw-Hill Publishers. (2013).
5. B.L. Tembe, M.S. Krishnan and Kamaluddin. Engineering Chemistry (NPTEL Web Course).
6. Shashi Chawla. A Textbook of Engineering Chemistry. Dhanpat Rai & Co, New Delhi.(2013).

Pattern of Continuous Assessment

Test -I for the theory portions	:	15 marks
Test -II for the theory portions	:	15 marks
Assignment from the Theory	:	05 marks
Laboratory Record	:	05 marks
Viva-voce	:	05 marks
Attendance	:	05 marks
TOTAL	**:**	**50 marks**

The students are required to submit the laboratory record.

23-200-0203B DIGITAL ELECTRONICS

Course Outcomes:
On completion of this course the student will be able to:
1. Understand the fundamental boolean functions and basic building blocks of Digital systems
2. Design Optimal digital circuits using basic building blocks
3. Analyse Basic digital circuits
4. Understand HDL models of simple circuits

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1									1
CO2	2	2	3									
CO3	1	3	1	2								
CO4	1	1	1	1	3				1			2

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Digital Concepts and Techniques: Binary arithmetic, Binary coded Decimal, Excess - 3 code, Gray Code. Boolean algebra-Standard Sum of products, Standard Product of sums. Logic gates. Minimization of Boolean function :Karnaugh Map (up to 5 variables) and Quine - McClusky methods. Variable entered mapping. Design of optimal logic function from a given problem statement.
HDL: Basic concepts and modelling of simple circuits.

Module II

Combinational circuits: Half adder, Full adder, Subtractor, Ripple Carry adder, Carry look ahead adder, BCD adder, multiplexer, demultiplexer, Basic decoder and encoder circuits,Binary Multiplication.
Sequential circuits: Flip-flops – (RS /JK / MS/T / D)
Serial Adder-Difference between Parallel Adder and Serial Adder
HDL: Models of simple combinational circuits

Module III

Shift Registers: various types - Counters : Asynchronous and synchronous counters, Up-Down counter, Shift Register Counters - Sequence generators
HDL: Models of simple sequential circuits.

Module IV

Implementation of logic functions using PLA, PROM. Error Detection and Correction: Parity, (7,4) Hamming code. Practical design considerations: Logic families- Standard logic levels- Current And Voltage Parameters- fan in and fan out-Propagation delay, Noise margin, Speed power product, setup time, hold time.
TTL family NAND gate working principle, Totem pole configuration- Transfer characteristics, Tri-state logic gate.
***Note**: HDL portion of each module to be evaluated based on assignments ONLY, as part of Continuous Evaluation, subject to a maximum weightage of 50% of marks allocated for assignments*

References:

1. Floyd, Thomas L. Digital fundamentals, 11/e. Pearson Education India, 2017. 978-9332584600
2. Kumar, A. Anand. Fundamentals of digital circuits.4/e PHI Learning, 2016. 978-8120352681
3. Stephan Brown & Zvonko Vranesic, Fundamentals of Digital Logic with Verilog Design, 2/e, McGraw-Hill, 2007, ISBN-13: 978-0077211646
4. Taub, Herbert, and Donald L. Schilling. Digital integrated electronics. McGraw Hill, India, 2016 978-0070265080
5. Roth Jr, Charles H., Larry L. Kinney, and Eugene B. John. Fundamentals of logic design. Cengage Learning, 2020. 978-9353502645

23-200-0204B OBJECT ORIENTED PROGRAMMING IN C++

Course Outcomes:

On completion of this course the student will be able to:

1. Understand the basic concepts of Object-Oriented Programming.
2. Describe the object-oriented paradigm with concepts of streams, classes, functions, data and objects.
3. Implement object-oriented programming constructs like encapsulation, inheritance and polymorphism.
4. Understand dynamic memory management techniques using pointers, constructors, destructors, etc.
5. Identify classes including data, methods and the relationship among the classes from a given problem statement and solve the problem using object-oriented constructs in C++.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											1
CO2	3	2										1
CO3	3	2	3									2
CO4	3											1
CO5	3	3	3		2	2			2			2

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Procedure oriented programming, Object oriented programming paradigm, Basic concepts of object-oriented programming, Benefits of OOP, Introduction to C++ programming, data types, variables, control statements (if, if else, switch), iteration (for, while, do...while). Console I/O operations - formatted and unformatted –managing output with manipulators. Functions in C++, call and return by reference, inline functions, default arguments, const arguments.

Module II

Classes and objects, specifying a class, defining member functions, Memory allocation for objects Static data members, Static member functions, Arrays of objects, const member functions Constructors and Destructors, Constructors: default, parameterised, with default arguments, copy constructor, destructors, Friend functions.
Introduction to pointers, new and delete operators, Pointers to objects, this pointer.

Module III

Inheritance: Defining derived classes, Single inheritance, Multilevel inheritance, multiple inheritance, Hierarchical inheritance, Hybrid inheritance, Virtual base classes, Abstract classes, Constructors in derived classes .
Polymorphism: Function overloading, operator overloading: overloading unary operators, overloading binary operators, overloading binary operators using friends, manipulation of strings using operators, Type conversions: basic to class, class to basic, class to class.

Module IV

Pointers to derived classes, virtual functions, pure virtual functions.
Working with files: classes for fstream operations, opening and closing of file, detecting end of file, file modes, file pointers and manipulators, sequential input and output operations, random access, Templates, Exception handling.

List of programs to practice:

1. Implementation of classes and objects.
2. Implementation of constructors and constructor overloading.
3. Implementation of methods and method overloading.
4. Implementation of different types of inheritance: Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Multiple Inheritance, Hybrid Inheritance.
5. Implementation of polymorphism.
6. Implementation of File handling
7. Assignment: Design any real time application using object oriented concepts and develop the solution using the C++programming language. For this the students can form a project team with a maximum 4 members per team.The team can select a socially relevant problem from various domains such as health, safety, education, agriculture, legal etc. At the end of the semester, the team has to demonstrate their product and submit a report. The team will be assessed through rubrics.

References:

1. Balagurusamy, E. (2020). Object oriented programming with C++ (8th ed.).Tata McGraw Hill. New Delhi.
2. Lafore, R., & Lafore, R. (2002). Object oriented programming in C++ (4th ed.).Sams Pub.Indianapolis, Indiana.
3. Stroustrup, B. (2013). The C++ programming language (4th ed.). Reading, Mass.:Addison Wesley.
4. Kamthane, A. (2003). Object oriented programming with ANSI and Turbo C++.Pearson Education.Delhi, India.
5. Schildt, H. (2012). C++ the complete reference (5th ed.). Osborne McGraw Hill. Berkeley.

Pattern of Continuous Assessment

Test -I for the theory portions	:	15 marks
Test -II for the theory portions	:	15 marks
Assignment from the Theory	:	05 marks
Assignment from the Practice	:	10 marks
Attendance	:	05 marks
TOTAL	**:**	**50 marks**

The students are required to submit the practice record

23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS

Course Outcomes:

On completion of this course the student will be able to:

1. Understand the features & components of Cyber Physical Systems
2. Understand the elementary constructs of Arduino Software
3. Develop optimal programs & circuits for interfacing various sensors with Arduino
4. Apply Arduino IDE for developing suitable programs for interfacing sensors & actuators with Arduino & Node MCU

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2									1
CO2	1	2	2									
CO3	1	2	2	2								1
CO4	1	1	1	1	2							1

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Introduction to CPS, Key features of CPS-reactive computation, concurrency, Feedback Control of the Physical World, real time computation, safety critical applications, general structure of a CPS, Examples of CPS.
IoT- Characteristics of IoT, Enabling Technologies, Concept of Transducers/sensors-Primary & Secondary, active & passive, analog & digital, Concept about actuators-thermal, electric & mechanical, IoT stack, Levels of IoT.

Module II

Arduino Basics: Development Boards-Arduino Uno, Node MCU, Arduino IDE, General Program Structure, Basic data types, variables & constants, Operators, control statements, loops, functions, time functions, arrays.

Module III

Arduino-I/O functions, Arduino PWM, Arduino Communication, LED Interfacing-Sinking & Sourcing methods of LED Connection, LED blinking, fading, Analog read, LED bar graph display, Seven Segment LED Display, interfacing sensors with Arduino-humidity, temperature, water detector, PIR, ultrasonic sensor, LDR, Interfacing Push Button Switch-Pull Up & Pull-Down Connection.

Module IV

Arduino for motor control- Control of DC Motor, Servo motor & Stepper Motor.
Introduction to NodeMCU/ESP32, Overview of NodeMCU and its features, Programming NodeMCU via Arduino IDE, Interfacing LED, Gas Sensor, Introduction to Wi-Fi Connectivity with Node MCU.

References:

1. Rajeev Alur. Principles of Cyber-Physical Systems, MIT Press 2015
2. Edward Ashford Lee, Sanjit Arunkumar Seshia. Introduction to Embedded Systems-A Cyber-Physical Systems Approach -MIT Press 2017
3. Shriram K Vasudevan, Abhishek S Nagarajan, RMD Sundaram.Internet of Things, WILEY 2020.
4. Srinivasa K.G, Siddesh G.M, Hanumantha Raju R.Internet of Things, CENGAGE, 2018.
5. Arduino-Tutorials point

23-200-0206B ENVIRONMENTAL AND LIFE SCIENCES

Course Outcomes:
On completion of this course the student will be able to:

1. Identify the global environmental issues
2. Examine the types of pollution in society along with their sources
3. Elucidate the basic biological concepts via relevant industrial applications and case studies.
4. Evaluate the principles of design and development, for exploring novel bioengineering projects.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2				3	3					
CO2	2	2				2	3					
CO3	2	2				2	2					
CO4	2	2				3	3					

1-Slightly**; 2-**Moderately**; 3-**Substantially

Module I

Environment, Ecosystems and Biodiversity: Definition, scope and importance of environment
— need for public awareness — concept of an ecosystem — structure and function of an ecosystem — producers, consumers and decomposers — energy flow in the ecosystem — ecological succession — food chains, food webs and ecological pyramids — Introduction, types, characteristic features, structure and function of the (a) forest ecosystem (b) grassland ecosystem (c) desert ecosystem (d) aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) — Introduction to biodiversity definition: genetic, species and ecosystem diversity —biogeographical classification of India — value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values — Biodiversity at global, national and local levels — India as a mega-diversity nation — hot-spots of biodiversity — threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts — endangered and endemic species of India — conservation of biodiversity: In-situ and ex-situ conservation of biodiversity. Field study of common plants, insects, birds; Field study of simple ecosystems — pond, river, hill slopes, etc.

Module II

Natural Resources: Forest resources: Use and over-exploitation, deforestation, case studies- timber extraction, mining, dams and their effects on forests and tribal people — Water resources: Use and over- utilization of surface and ground water, floods, drought, conflicts over water, dams-benefits and problems — Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies — Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies — Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy sources. case studies — Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification — role of an individual in conservation of natural resources. The concept of sustainable development.

Module III

Biomolecules and their Applications (Qualitative): Carbohydrates (cellulose-based water filters, PHA and PLA as bioplastics), Nucleic acids (DNA Vaccine for Rabies and RNA vaccines for Covid19, Forensics – DNA fingerprinting), Proteins (Proteins as food – whey protein and meat analogs, Plant based proteins), Lipids (biodiesel, cleaning agents/detergents), Enzymes (glucose-oxidase in biosensors, Ligninolytic enzyme in bio-bleaching).
Nature-Bioinspired Materials and Mechanisms (Qualitative): Echolocation (ultrasonography, sonars), Photosynthesis (photovoltaic cells, bionic leaf). Bird flying (GPS and aircrafts), Lotus leaf effect (Super hydrophobic and self-cleaning surfaces), Plant burrs (Velcro), Shark skin (Friction reducing swimsuits), Kingfisher beak (Bullet train). Human Blood substitutes - hemoglobin-based oxygen carriers (HBOCs) and perfluorocarbons (PFCs).

Module IV

Human Organ Systems and Bio Designs (Qualitative): Brain as a CPU system (architecture, CNS and Peripheral Nervous System, signal transmission, EEG, Robotic arms for prosthetics. Engineering solutions for Parkinson’s disease). Eye as a Camera system (architecture of rod and cone cells, optical corrections, cataract, lens materials, bionic eye). Heart as a pump system (architecture, electrical signaling – ECG monitoring and heart related issues, reasons for blockages of blood vessels, design of stents, pace makers, defibrillators). Lungs as purification systems (architecture, gas exchange

mechanisms, spirometry, abnormal lung physiology - COPD, Ventilators, Heart-lung machine). Kidney as a filtration system (architecture, mechanism of filtration, CKD, dialysis systems). Muscular and Skeletal Systems as scaffolds (architecture, mechanisms, bioengineering solutions for muscular dystrophy and osteoporosis).
Bioprinting techniques and materials, 3D printing of ear, bone and skin. 3D printed foods.

References:

1. Rajagopalan, R. Environmental Studies: From Crisis to Cure. Oxford University Press, New Delhi, (2015).
2. Erach Bharucha. Textbook of Environmental Studies and Ethics. Universities Press (India), Hyderabad, (2013).
3. Thyagarajan S., Velmurugan N., Rajesh M.P., Nazeer R.A., Thilagaraj W., Barathi S., and Jagannathan M.K. Biology for Engineers, Tata McGraw-Hill, New Delhi, (2012).
4. Arthur T. Johnson. Biology for Engineers, CRC Press, Taylor and Francis, (2019).
5. Sohini Singh and Tanu Allen. Biology for Engineers, Vayu Education of India, New Delhi, (2020).
6. Ibrahim Ozbolat. 3D Bioprinting: Fundamentals, Principles and Applications, Academic Press, (2016).

23-200-0207B DIGITAL ELECTRONICS LAB

Course Outcomes:

On completion of this course the student will be able to:

1. Understand working of gates, flip flops MUX, DEMUX, Shift registers ,counters etc.
2. Design digital circuits using appropriate ICs
3. Understand the timing diagrams
4. Develop teamwork skills

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1									1
CO2	2	2	3									
CO3	1	3	1	2								
CO4	1	1	1	1	3				1			2

1-Slightly**; 2-**Moderately**; 3-**Substantially

Experiments:

Introduction to Data sheet of various digital ICs and their familiarization to be given before the hands on sessions

1. Half adder and full adder using standard logic gates / NAND gates.
2. Code converters - Binary to Gray and gray to Binary with mode control
3. Binary addition and subtraction (a) 1's complement (b) 2's complement(using7483)
4. BCD adder using7483.
5. Study of MUX, DEMUX & Decoder Circuits and ICs
6. Set up R-S JK & JK Master slave flip flops using NAND/NOR Gates
7. Asynchronous UP / DOWN counter using JK Flipflops
8. Design and realization of sequence generators.
9. Study of shift registers and Implementation of Johnson and Ring counters using them .
10. Study of counter ICs 7490, 7492, 7493.
11. Study of seven segment display and decoder driver (7447)- virtual lab

Hands-on wiring experiments may be supplemented by simulation using CAD tools / virtual labs etc.
At least 8 experiments must be mandatorily completed by every student and recorded.
Students are required to submit a simple project fully conceived, designed and developed by them at the end of the semester

References:

1. Herbert Taub, Donald Schilling, Digital Integrated Electronics, Tata Mc Graw Hill, 1/e, (2008), ISBN:9780070265080

23-200-0208B BASIC ELECTRONICS LAB

Course Outcomes:

On completion of this course the student will be able to:

1. To design and implement simple hardware circuits using electronic devices and digital ICs and to test the performance and its applications.
2. To use the basic logic gates and various reduction techniques of digital logic circuit in detail.
3. To design simple circuits and mini projects (groups) using sensors and electronic components

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1									1
CO2	1	2	1									
CO3	1	3	1	2					1			

1-Slightly**; 2-**Moderately**; 3-**Substantially

Experiments:

1. Familiarization of electronic components and Electronic instruments – Power Supply, Function Generator, CRO, Multimeter.
2. VI characteristics of PN junction diode.
3. Clipping and clamping circuits
4. Design Rectifying circuits: (with and without filter)
    a. Half Wave Rectifier
    b. Full Wave Rectifier
5. Characterization of Passive Integrator and Differentiator Circuits.
6. Characterization of Transistor CE Configuration.
7. Design CE Amplifier for a particular Gain.
8. Electronic Systems Hardware Familiarization.
9. Introduction to PCB design.

References:

1. Jacob Milman & Christos C Halkias, Integrated Electronics: Analog and Digital Circuits and Systems, McGraw Hill Education, 2 nd edition (2017).
2. David M. Buchla, Thomas L. Floyd, Electronics Pearson Education Limited, Year: 2014.